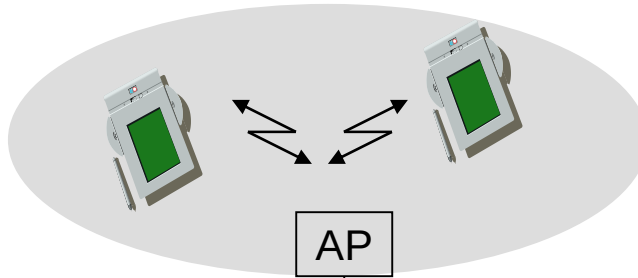


IEEE 802.11

José Augusto Afonso
Jose.afonso@dei.uminho.pt

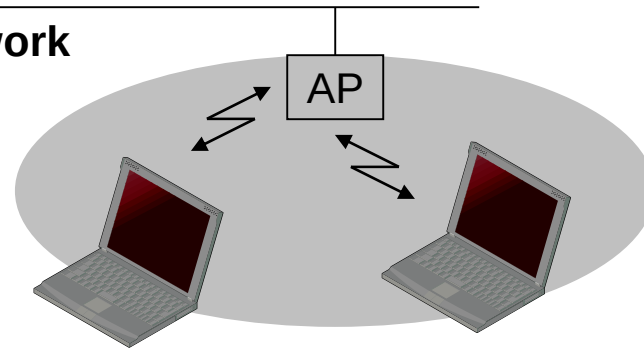
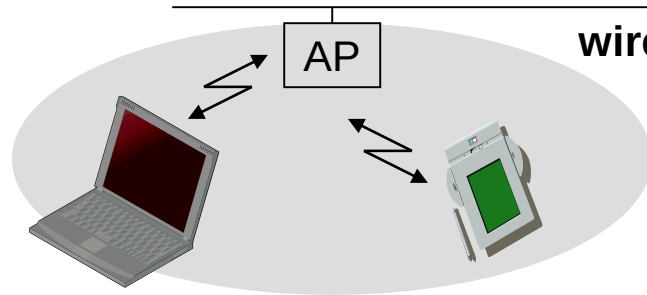
Infrastructure vs. Ad-hoc Networks

infrastructure
network

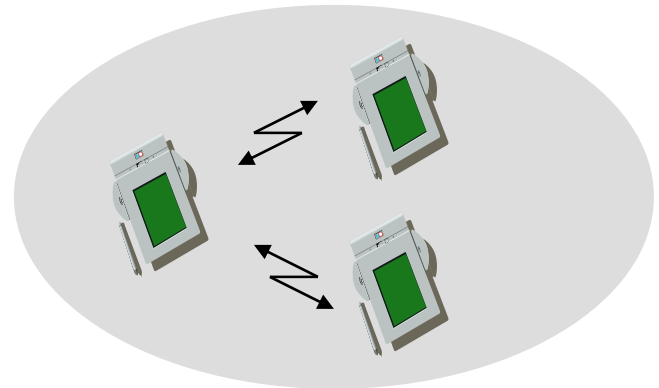
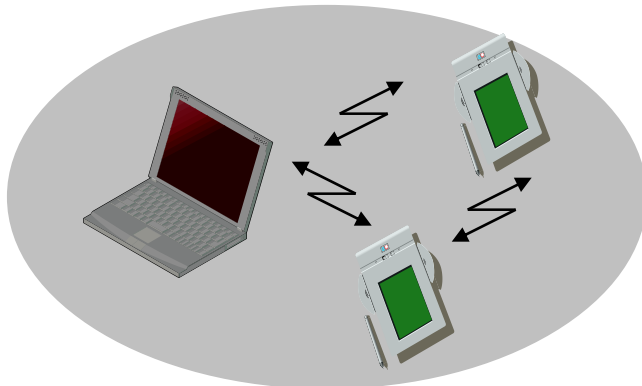


AP: Access Point

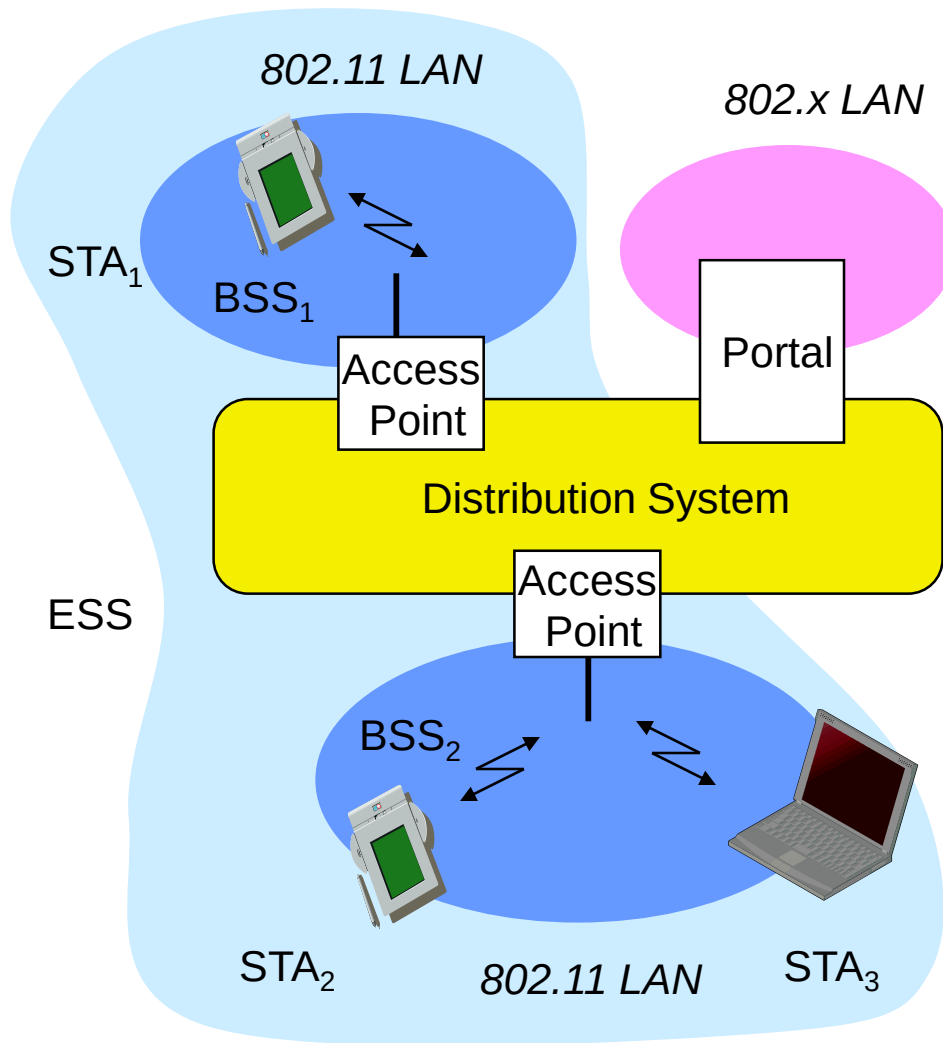
wired network



ad-hoc network



802.11 - Architecture of an Infrastructure Network



* Schiller, 2008

Station (STA)

- terminal with access mechanisms to the wireless medium and radio contact to the access point

Basic Service Set (BSS)

- group of stations using the same radio frequency

Access Point

- station integrated into the wireless LAN and the distribution system

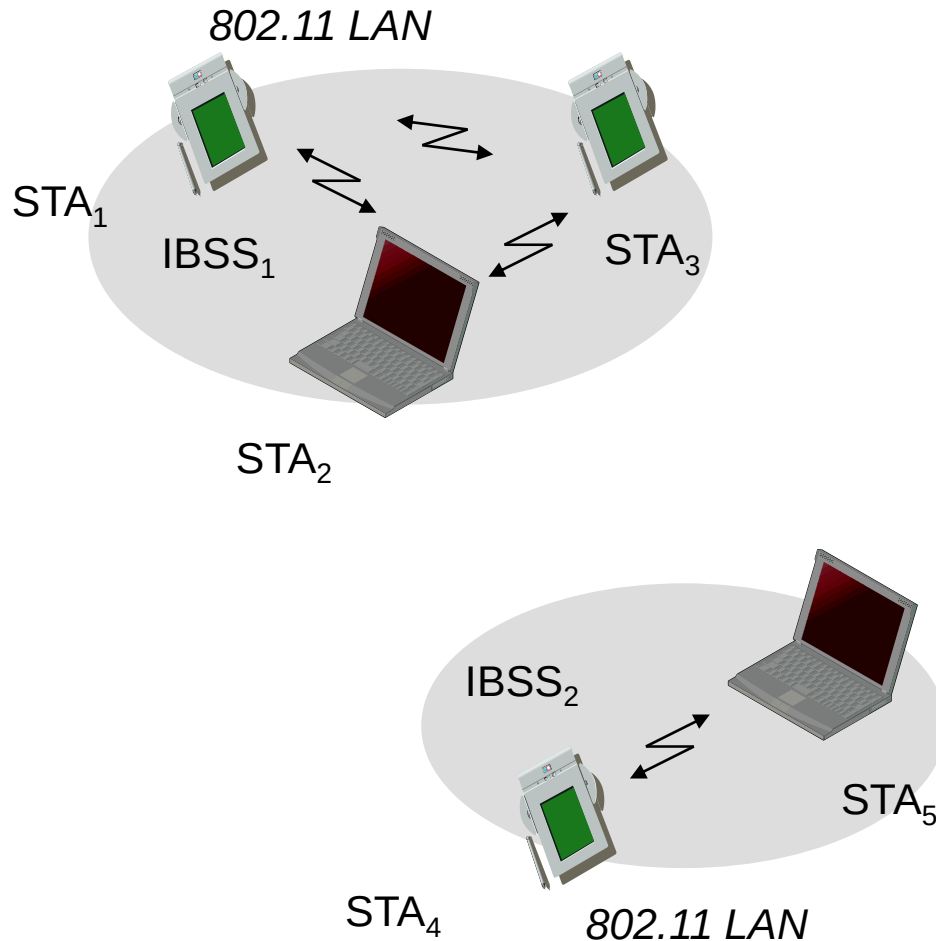
Distribution System

- interconnection of APs to form one logical network (ESS: Extended Service Set) based on several BSS

Portal

- bridge to other (wired) networks

802.11 - Architecture of an Ad-hoc Network



Direct communication within a limited range

- Station (STA): terminal with access mechanisms to the wireless medium
- Independent Basic Service Set (IBSS): group of stations using the same radio frequency

Physical Layer (Classical)

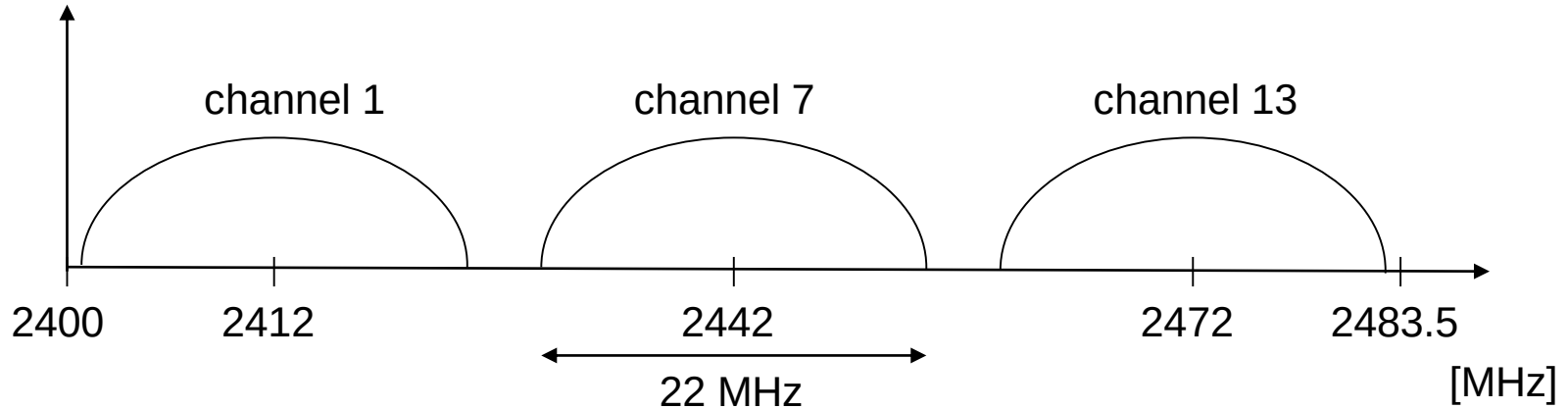
- 3 versions: 2 radio (2.4 GHz band), 1 IR
 - introduced in 1997
 - data rates 1 or 2 Mbit/s
- FHSS (Frequency Hopping Spread Spectrum)
 - spreading, despreading, signal strength, typ. 1 Mbit/s
 - min. 2.5 frequency hops/s (USA), two-level GFSK modulation
- DSSS (Direct Sequence Spread Spectrum)
 - DBPSK modulation for 1 Mbit/s (Differential Binary Phase Shift Keying), DQPSK for 2 Mbit/s (Differential Quadrature PSK)
 - preamble and header of a frame is always transmitted with 1 Mbit/s, rest of transmission 1 or 2 Mbit/s
 - chipping sequence: +1, -1, +1, +1, -1, +1, +1, +1, -1, -1, -1 (Barker code)
- Infrared
 - 850-950 nm, diffuse light, typ. 10 m range

Physical Layer

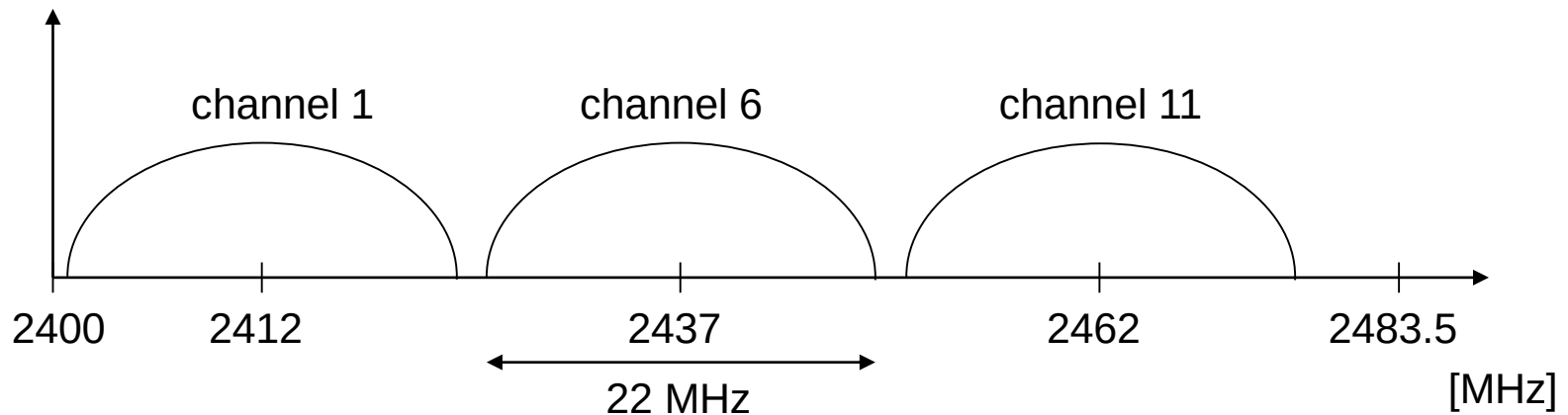
- 2.4 GHz band
 - DSSS (*Direct Sequence Spread Spectrum*) - 802.11
 - CCK (*Complementary Code Keying*) - 802.11b
 - OFDM (*Orthogonal Frequency Division Multiplexing*) - 802.11g
 - MIMO (*Multiple-Input Multiple-Output*) - 802.11n
 - FHSS (*Frequency Hopping Spread Spectrum*) - 802.11
- 5 GHz band
 - OFDM - 802.11a
 - MIMO - 802.11n
- Infrared band
 - DFIR (*Diffuse Infrared*) - 802.11

2.4 GHz Band Channels

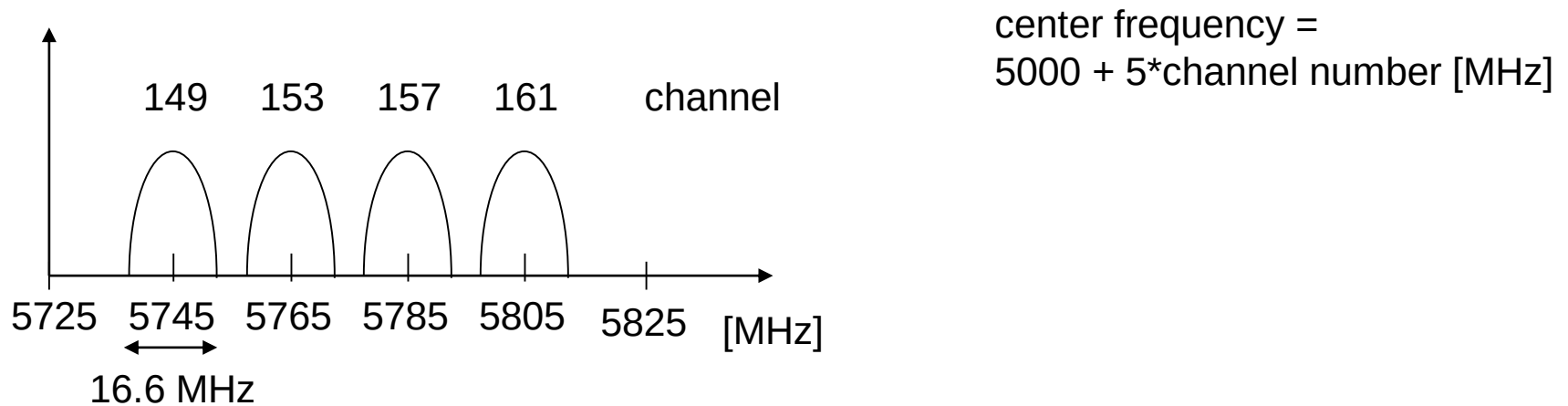
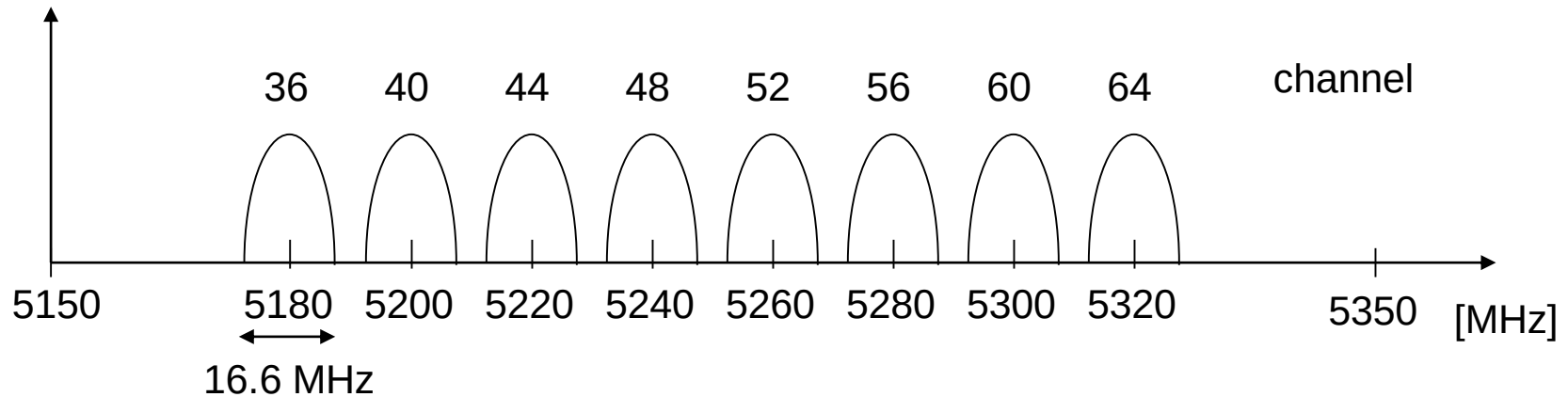
Europe (ETSI)



US (FCC)/Canada (IC)



5 GHz Band Channels (US-NII)



802.11 DSSS PHY Frame Format

Synchronization

- synch., gain setting, energy detection, frequency offset compensation

SFD (Start Frame Delimiter)

- 1111001110100000

Signal

- data rate of the payload (0A: 1 Mbit/s DBPSK; 14: 2 Mbit/s DQPSK)

Service

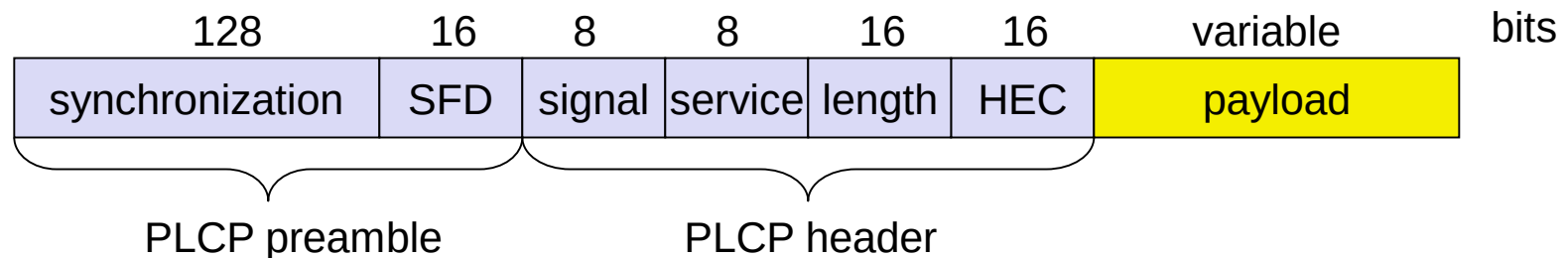
- future use, 00: 802.11 compliant

Length

- length of the payload

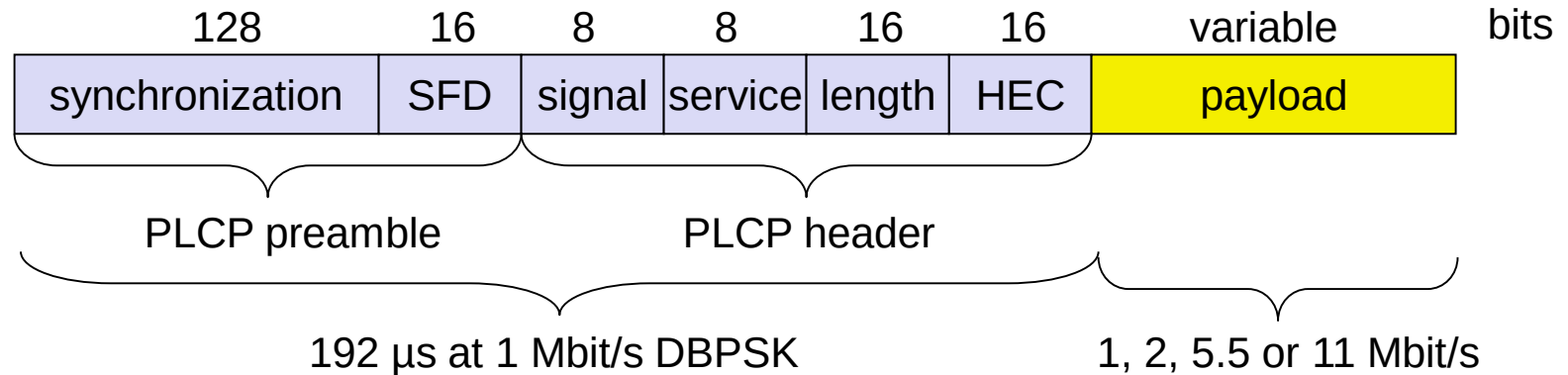
HEC (Header Error Check)

- protection of signal, service and length, $x^{16}+x^{12}+x^5+1$

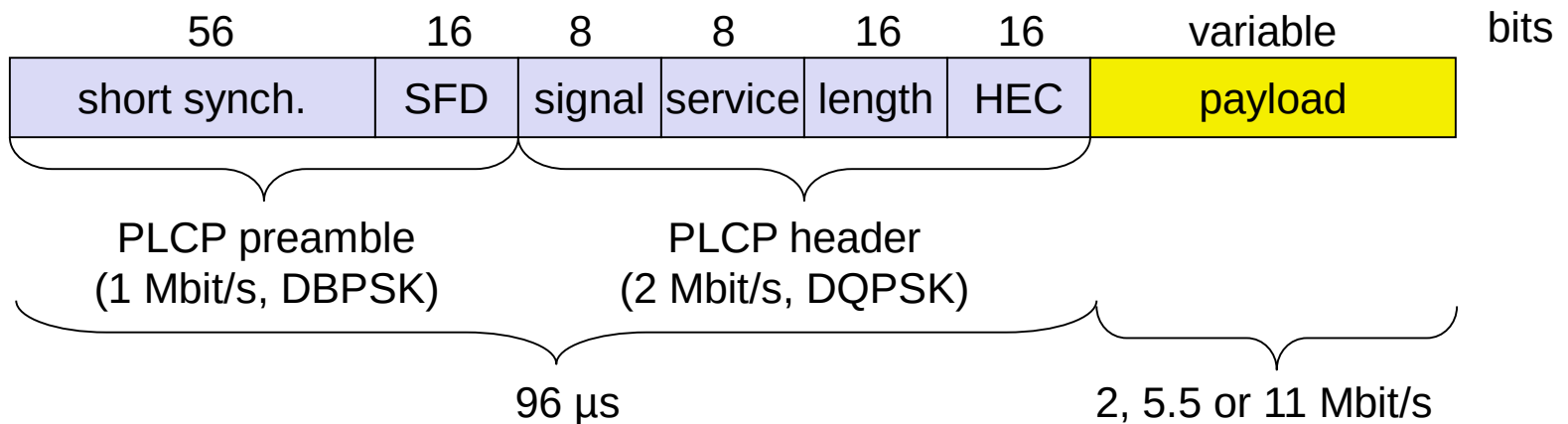


IEEE 802.11b – PHY Frame Formats

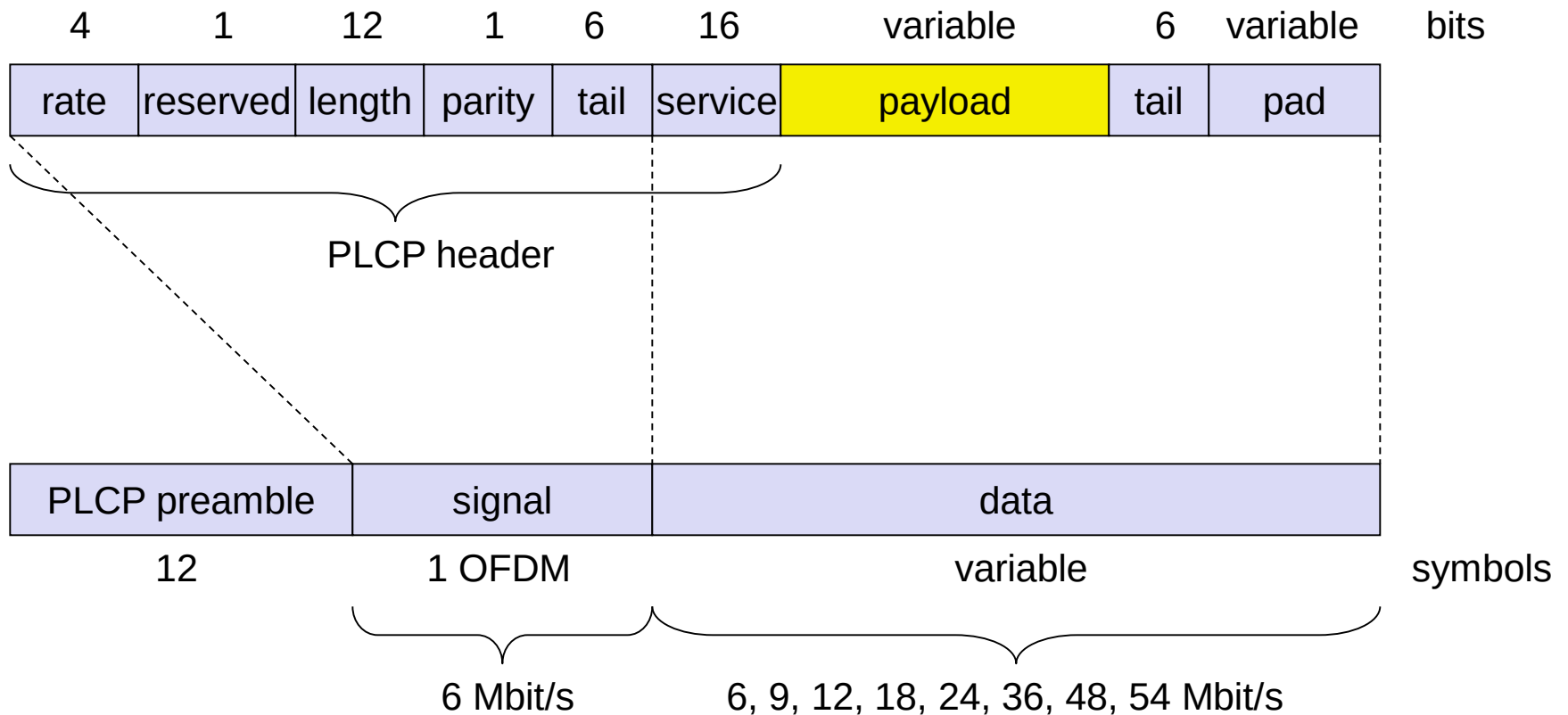
Long PLCP PDU format



Short PLCP PDU format (optional)

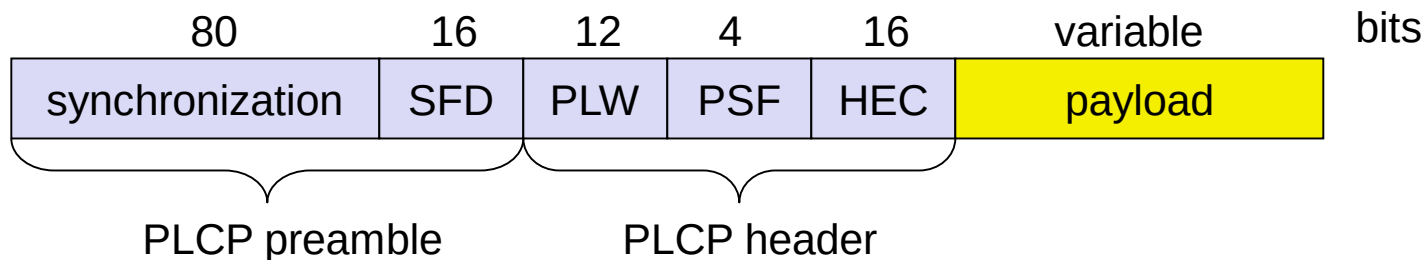


IEEE 802.11a – PHY Frame Format



802.11 FHSS PHY Packet Format

- Synchronization
 - synch with 010101... pattern
- SFD (Start Frame Delimiter)
 - 0000110010111101 start pattern
- PLW (PLCP_PDU Length Word)
 - length of payload incl. 32 bit CRC of payload, $PLW < 4096$
- PSF (PLCP Signaling Field)
 - data rate of payload (1 or 2 Mbit/s)
- HEC (Header Error Check)
 - CRC with $x^{16}+x^{12}+x^5+1$



Medium Access Control

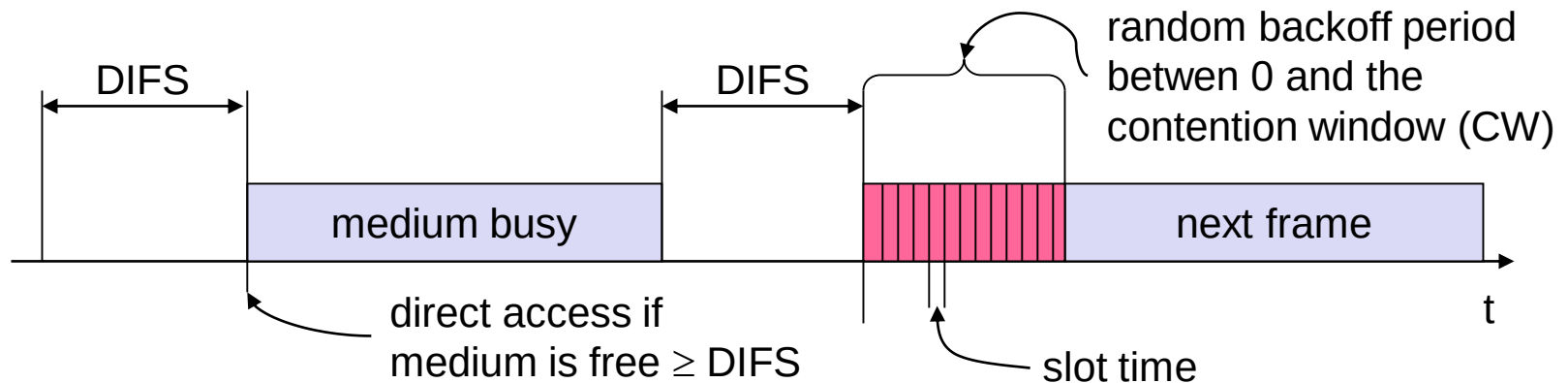
- Traffic services
 - Asynchronous Data Service (mandatory)
 - exchange of data packets based on “best-effort”
 - support of broadcast and multicast
 - implemented using DCF (Distributed Coordination Function)
 - Time-Bounded Service (optional)
 - implemented using PCF (Point Coordination Function)
- Access methods
 - DCF CSMA/CA (mandatory)
 - collision avoidance via randomized backoff mechanism
 - minimum distance between consecutive packets
 - ACK packet for acknowledgements (not for broadcasts)
 - DCF w/ RTS/CTS (optional)
 - Distributed Foundation Wireless MAC
 - avoids hidden terminal problem
 - PCF (optional)
 - access point polls terminals according to a list

MAC Layer Inter Frame Spaces

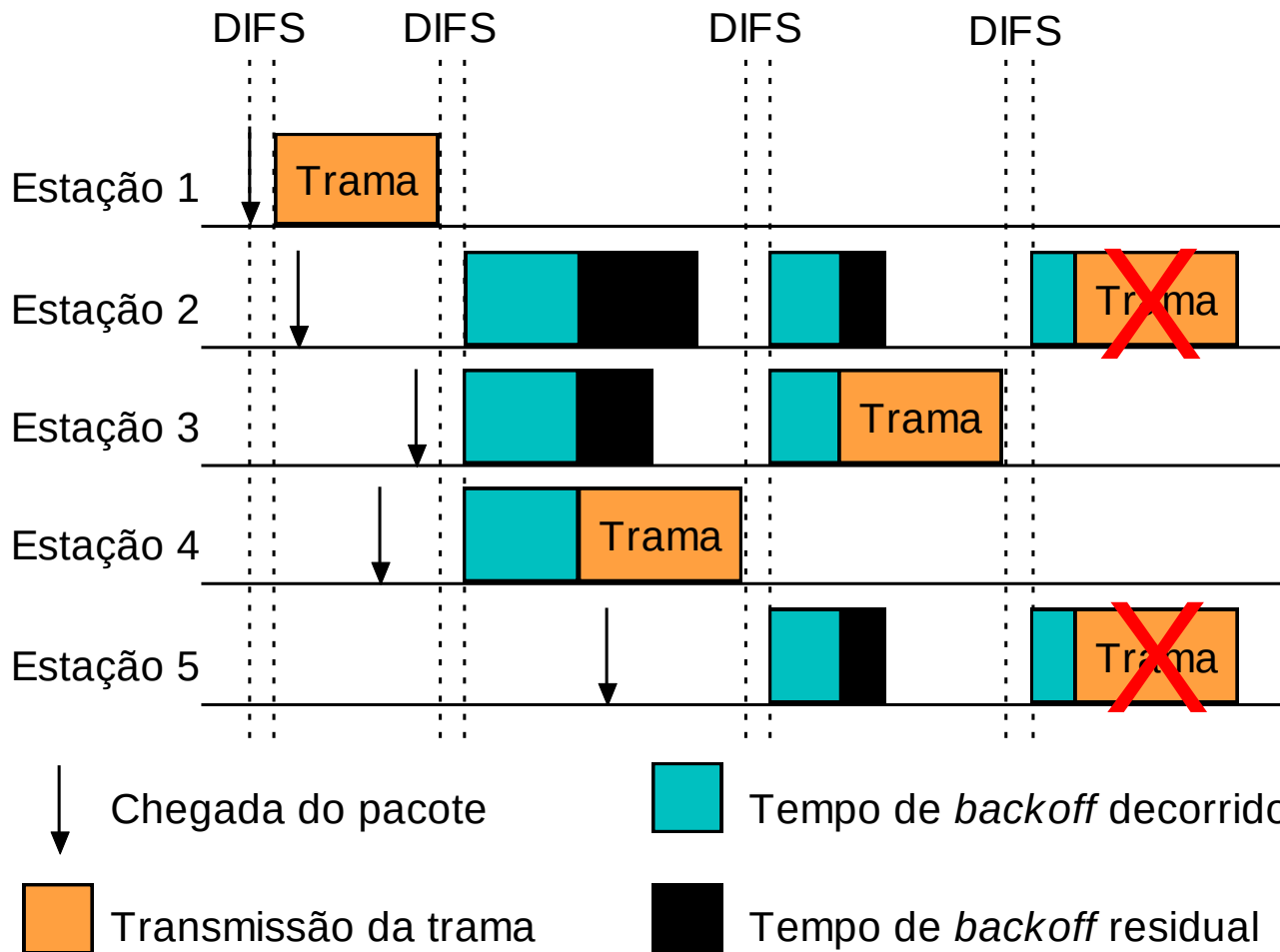
- Priorities
 - defined through different inter frame spaces
 - no guaranteed, hard priorities
 - SIFS (Short Inter Frame Spacing)
 - highest priority, for ACK, CTS, polling response
 - PIFS (PCF IFS)
 - medium priority, for time-bounded service using PCF
 - DIFS (DCF, Distributed Coordination Function IFS)
 - lowest priority, for asynchronous data service

CSMA/CA Access Method (DCF)

- A station ready to send starts sensing the medium (Carrier Sense based on CCA, Clear Channel Assessment)
- If the medium is free for the duration of an DCF Inter-Frame Space (DIFS), the station can start sending
- If the medium is busy, the station has to wait for a free DIFS, then the station must additionally wait a random backoff time (collision avoidance, multiple of slot-time)
- If another station occupies the medium during the backoff time of the station, the backoff timer stops



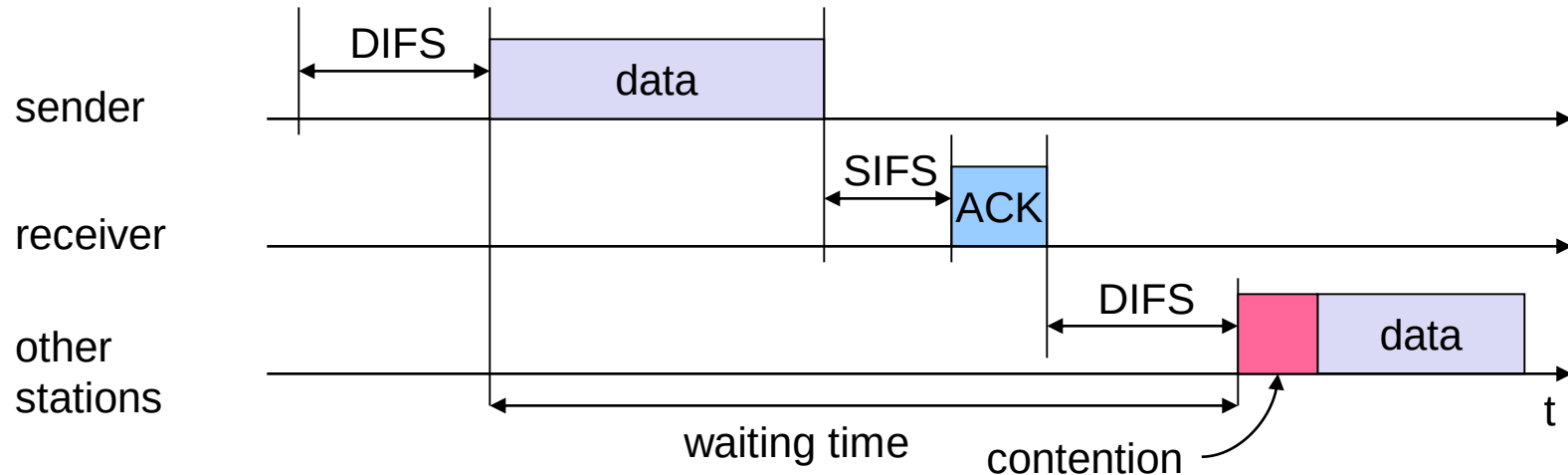
CSMA/CA Access Method



DIFS: Distributed Interframe Space

CW: Contention Window

Positive Acknowledgement Mechanism



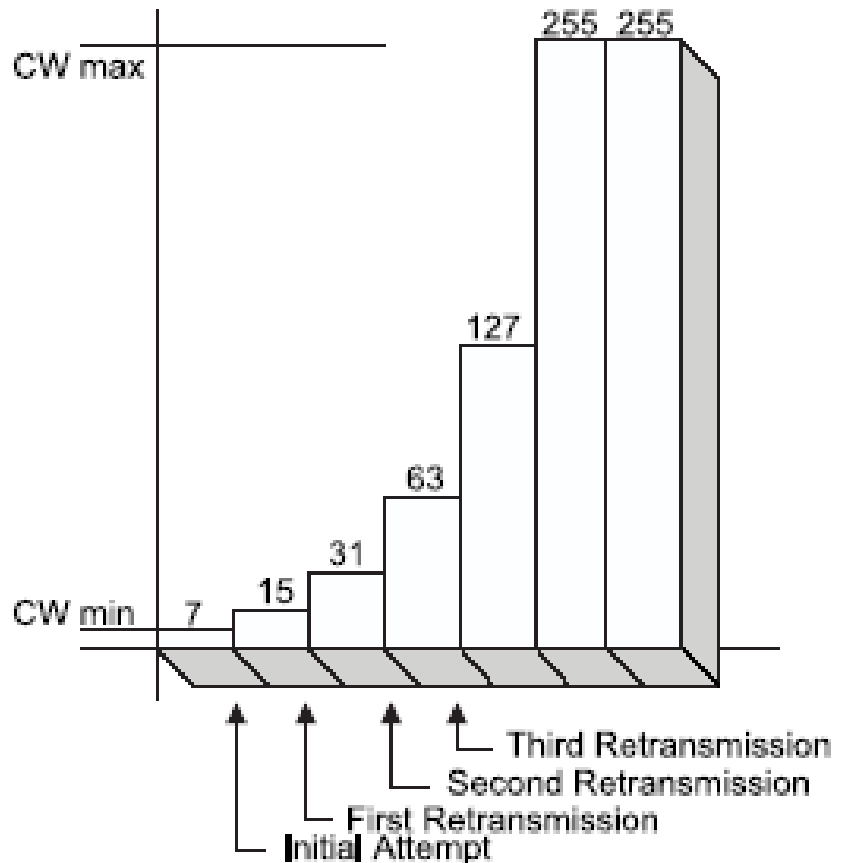
SIFS: Short Interframe Space

$$DIFS = SIFS + 2 \text{ SlotTime}$$

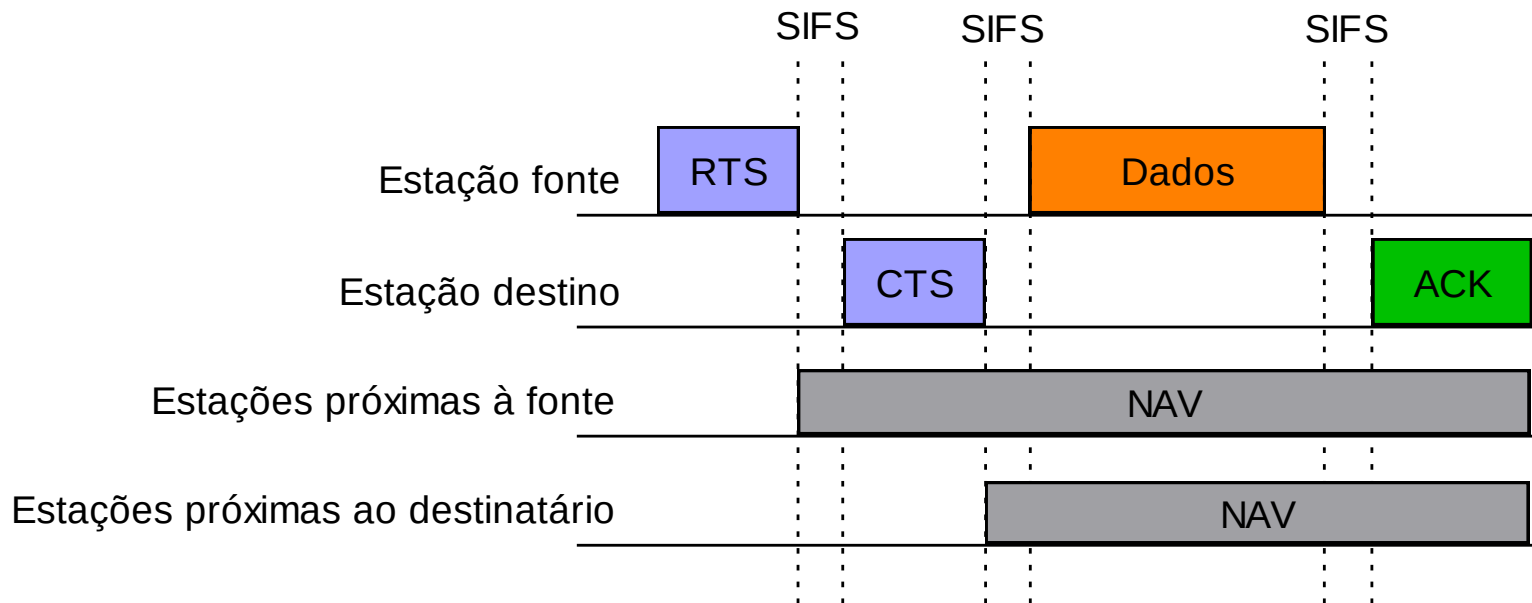
- As tramas de dados *unicast* recebidas sem erros (CRC OK) são reconhecidas pelo destinatário (trama ACK)
- Como $SIFS < DIFS$, as tramas ACK são transmitidas sem contenção
- Quando os dados são corrompidos, o destinatário não faz o reconhecimento negativo

Contention Window

- CW increases by power of 2 minus 1 by each retransmission attempt
- A successful transmission resets CW to CWmin
- Once CW reaches CWmax it remains at the value until a reset
- CWmin and CWmax are PHY specific



Mecanismo RTS/CTS



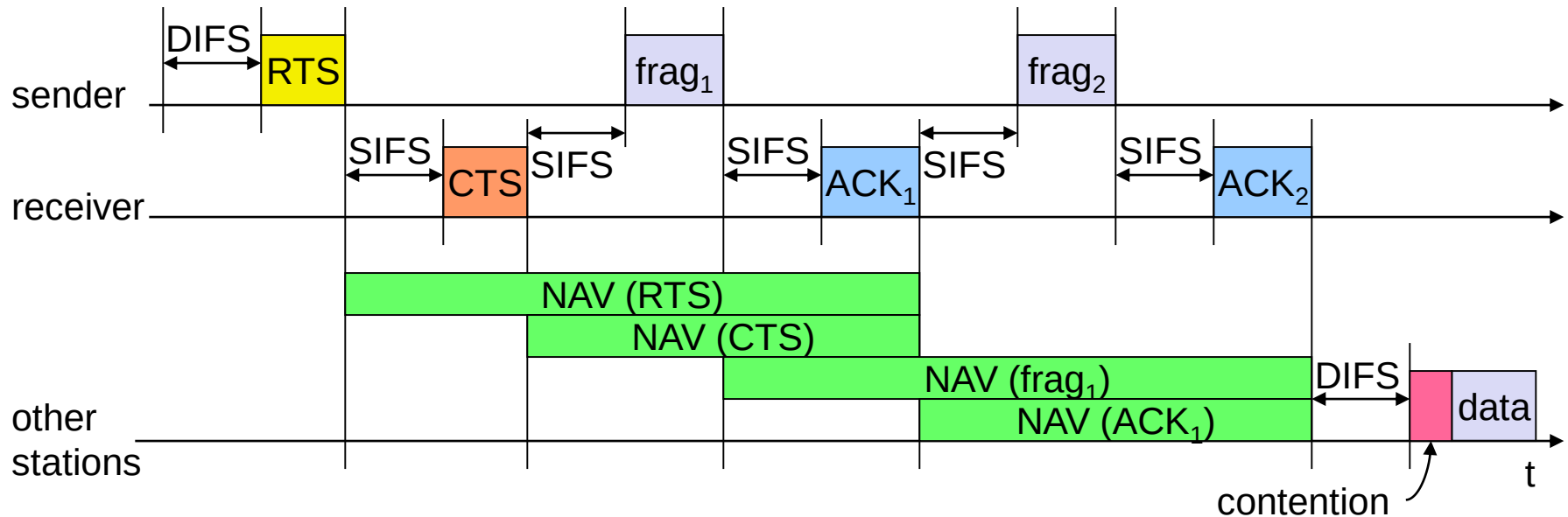
RTS: Request To Send

CTS: Clear To Send

NAV: Network Allocation Vector
(reserva do canal, portadora virtual)

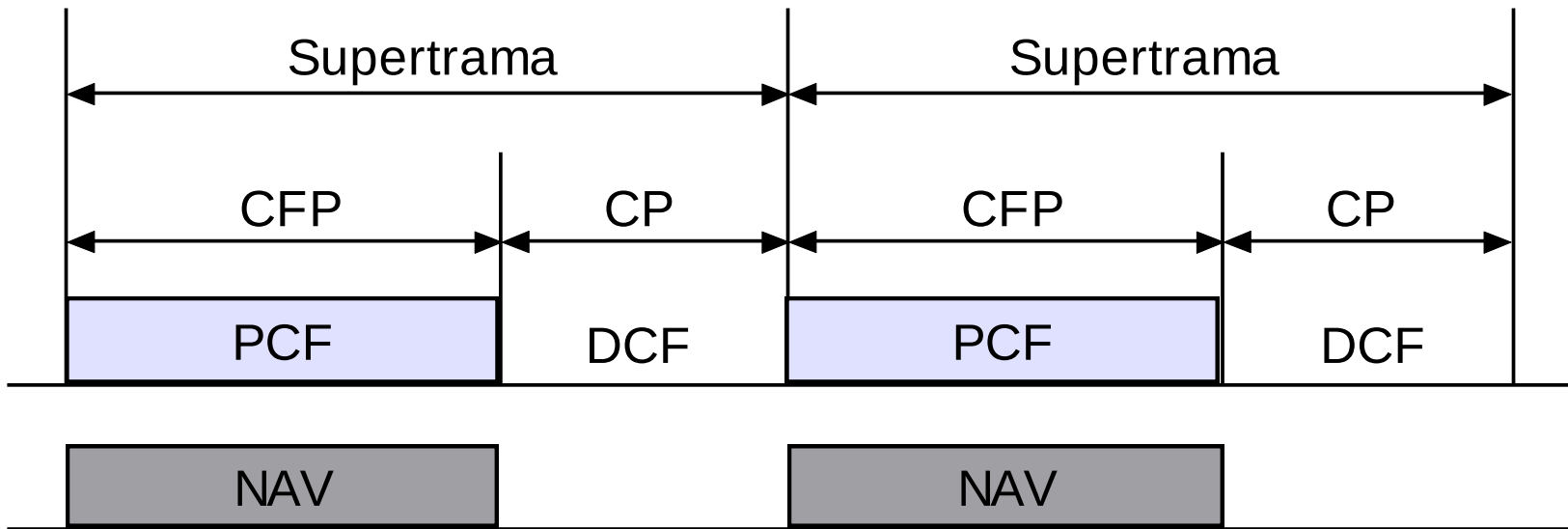
A utilização deste mecanismo depende do parâmetro *RTS Threshold*

Fragmentation Mechanism



- Creates MPDUs smaller than the original MSDU length
- Motivation
 - to increase reliability, by increasing the probability of successful transmission of the MSDU in cases where channel characteristics limit reception reliability for longer frames

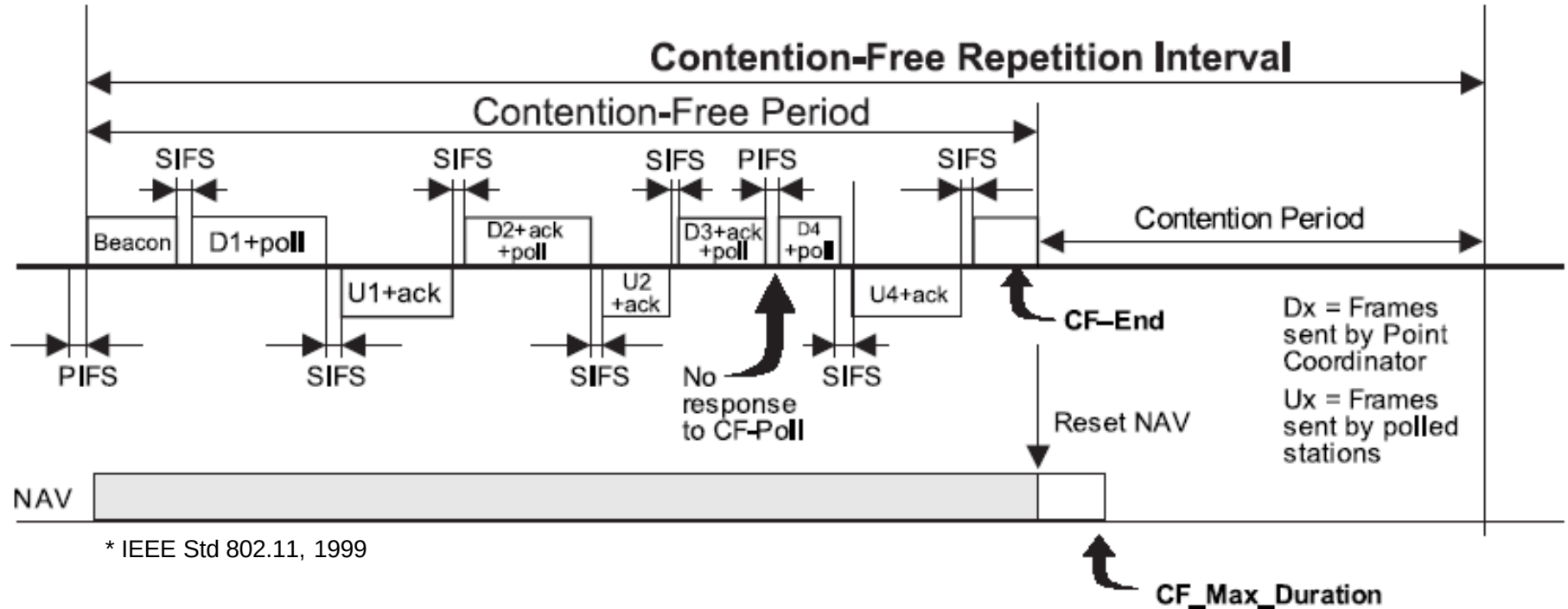
Point Coordination Function (PCF)



CFP: Contention Free Period

CP: Contention Period

Point Coordination Function (PCF)

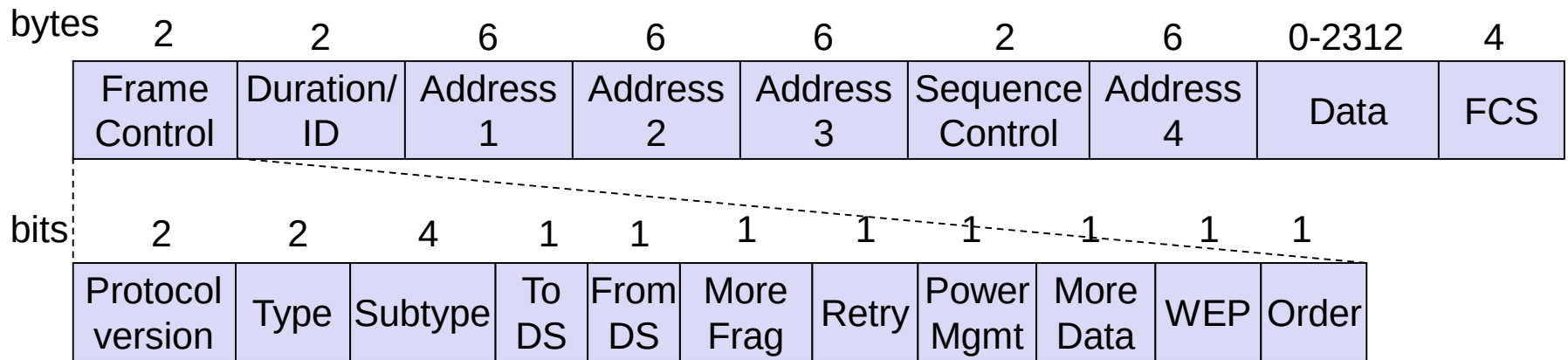


PIFS: PCF Interframe Space

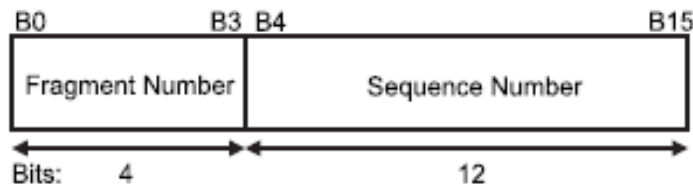
$$PIFS = SIFS + SlotTime$$

802.11 General MAC Frame Format

The fields Address 2, Address 3, Sequence Control, Address 4, and Frame Body are only present in certain frame types.



Sequence control field:



FCS: Frame Check Sequence

DS: Distribution System

WEPP: Wireless Equivalent Privacy

Frame Control Field

- **Protocol Version** (2 bits)
 - For this standard, the value of the protocol version is 0.
 - The revision level will be incremented only when a fundamental incompatibility exists between a new revision and the prior edition of the standard.
 - A device that receives a frame with a higher revision level than it supports will discard the frame.
- **Type** (2 bits) and **Subtype** (4 bits)
 - There are three frame types: control, data, and management.
 - Each of the frame types have several defined subtypes.
- **To DS** (1 bit)
 - Set to 1 in data type frames destined for the DS.
 - This includes all data type frames sent by STAs associated with an AP.
- **From DS** (1 bit)
 - Set to 1 in data type frames exiting the DS. Set to 0 in all other frames.

Frame Control Field

- **More Fragments** (1 bit)
 - Set to 1 in all data or management type frames that have another fragment of the current packet to follow and to 0 otherwise.
- **Retry** (1 bit)
 - Set to 1 in a frame that is a retransmission of an earlier frame.
 - Used to aid in the process of eliminating duplicate frames.
- **Power Management** (1 bit)
 - The value indicates the mode in which the station will be after the successful completion of a frame exchange sequence.
 - A value of 1 indicates that the STA will be in power-save mode.
 - A value of 0 indicates that the STA will be in active mode.
- **More Data** (1 bit)
 - Used to indicate that more packets are buffered for that station
- **WEP** (1 bit)
 - Indicates if the frame body is encrypted using the WEP algorithm

MAC Address Formats for Data Frames

scenario	to DS	from DS	address 1	address 2	address 3	address 4
ad-hoc network	0	0	DA	SA	BSSID	-
from infrastructure network	0	1	DA	BSSID	SA	-
to infrastructure network	1	0	BSSID	SA	DA	-
infrastructure network, within DS	1	1	RA	TA	DA	SA

BSSID: Basic Service Set Identifier

DA: Destination Address

SA: Source Address

RA: Receiver Address

TA: Transmitter Address

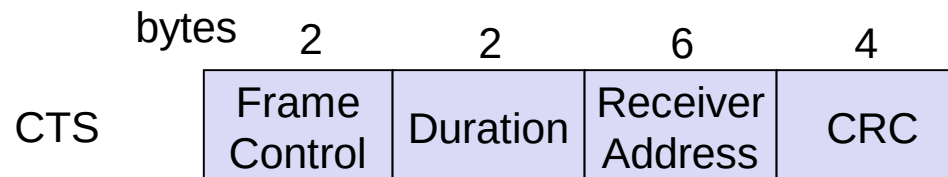
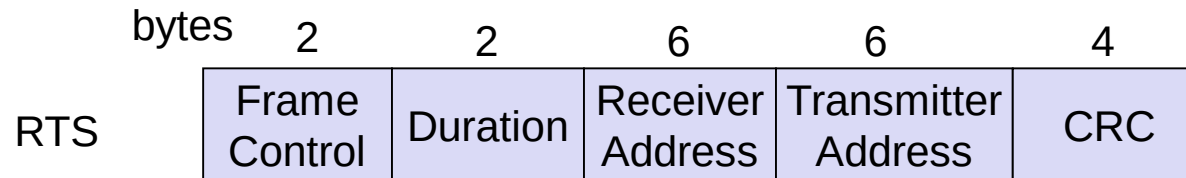
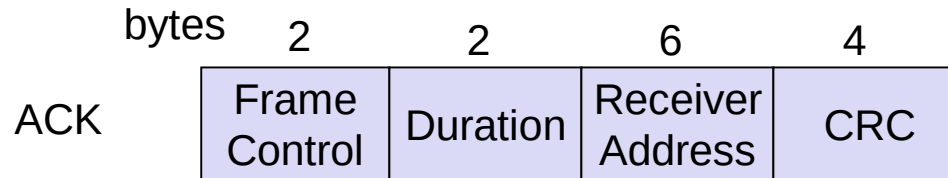
Data Frames

Type value b3 b2	Type description	Subtype value b7 b6 b5 b4	Subtype description
10	Data	0000	Data
10	Data	0001	Data + CF-Ack
10	Data	0010	Data + CF-Poll
10	Data	0011	Data + CF-Ack + CF-Poll
10	Data	0100	Null function (no data)
10	Data	0101	CF-Ack (no data)
10	Data	0110	CF-Poll (no data)
10	Data	0111	CF-Ack + CF-Poll (no data)
10	Data	1000–1111	Reserved
11	Reserved	0000–1111	Reserved

Control Frames

Type value b3 b2	Type description	Subtype value b7 b6 b5 b4	Subtype description
01	Control	0000–1001	Reserved
01	Control	1010	Power Save (PS)-Poll
01	Control	1011	Request To Send (RTS)
01	Control	1100	Clear To Send (CTS)
01	Control	1101	Acknowledgment (ACK)
01	Control	1110	Contention-Free (CF)-End
01	Control	1111	CF-End + CF-Ack

Control Frames: ACK, RTS, CTS



Management Frames

Type value b3 b2	Type description	Subtype value b7 b6 b5 b4	Subtype description
00	Management	0000	Association request
00	Management	0001	Association response
00	Management	0010	Reassociation request
00	Management	0011	Reassociation response
00	Management	0100	Probe request
00	Management	0101	Probe response
00	Management	0110–0111	Reserved
00	Management	1000	Beacon
00	Management	1001	Announcement traffic indication message (ATIM)
00	Management	1010	Disassociation
00	Management	1011	Authentication
00	Management	1100	Deauthentication
00	Management	1101–1111	Reserved

IEEE 802.11 Standards and Task Groups

- IEEE 802.11 - The original 1 Mbit/s and 2 Mbit/s, 2.4 GHz RF (DSSS and FHSS) and IR Wireless LAN standard (1999)
- IEEE 802.11a - 54 Mbit/s, 5 GHz standard (1999)
- IEEE 802.11b - Enhancements to 802.11 to support 5.5 and 11 Mbit/s (1999)
 - First version with widespread acceptance on the market
- IEEE 802.11e - QoS Enhancements (2005)
 - Improvements to the Media Access Control (MAC) layer
- IEEE 802.11g - 54 Mbit/s, 2.4 GHz standard (2003)
 - Successful successor of 802.11b, some performance loss during mixed operation with 11b

IEEE 802.11 Standards and Task Groups

- IEEE 802.11n - Higher throughput improvements (2009)
 - Adds support to MIMO antennas (Multiple Input Multiple Output)
 - Support to wider channels (40 MHz versus 20 MHz)
 - Allows data rates potentially up to 600Mbit/s
 - Operation on both the 2.4 GHz and the 5 GHz bands (optional)
- IEEE 802.11ac - Improvements over 802.11n (2013)
 - Wider channels (80 or 160 MHz versus 40 MHz) in the 5 GHz band
 - Higher-order modulation (up to 256-QAM vs. 64-QAM)

IEEE 802.11 Standards and Task Groups

- IEEE 802.11p - WAVE (Wireless Access in Vehicular Environments)
 - Communication between cars/roadside infrastructure and cars/cars
 - Support to Intelligent Transportation Systems (ITS) applications
 - Planned for relative speeds of min. 200 km/h and ranges over 1000 m
 - Usage of licensed 5.9 GHz band (5.850-5.925GHz) in North America
- IEEE 802.11ad - Operation on unlicensed 60 GHz band
 - New PHY layer for 60 GHz millimeter wave band (very different propagation characteristics compared to 2.4 and 5 GHz bands)
 - Backward compatibility with existing Wi-Fi devices and bands
 - Uses WiGig (Wireless Gigabit) brand name
 - Data rates of up to 7 Gbit/s

IEEE 802.11 Standards and Task Groups

- IEEE 802.11af - White-Fi (2014)
 - Operation in TV white space (unused) spectrum in the licensed VHF and UHF bands between 54 and 790 MHz
 - Lower frequency increases the possible range
 - Uses cognitive radio technology to limit interference for primary users, such as analog TV, digital TV, and wireless microphones (e.g., access points determine their position using GPS and query a database (GDB) to discover what channels are available)
 - Channels 6 to 8 MHz wide, up to four channels may be bonded
 - Up to 35.6 Mbit/s per used channel
- 802.11ah - Sub-1GHz unlicensed bands operation (2016)
 - Higher transmission range, less available bandwidth
 - Applications include large scale sensor networks and extended range hotspots